

MBBS IV Junior Clerkship (2019-2020)

Lectures 42

High white cell count:

Acute and Chronic Leukaemia

Bone Marrow Transplantation

Immunogenetics

Joycelyn Sim

Department of Medicine

Queen Mary Hospital

Dec 2023

Learning Objectives

1. **Leukaemia classification (WHO 2022)**
2. **General principles in managing patients with leukaemia**
3. Role of **haemtopoietic stem cell transplantation** in leukaemia treatment
4. Importance of **histocompatibility** in transplantation immunology

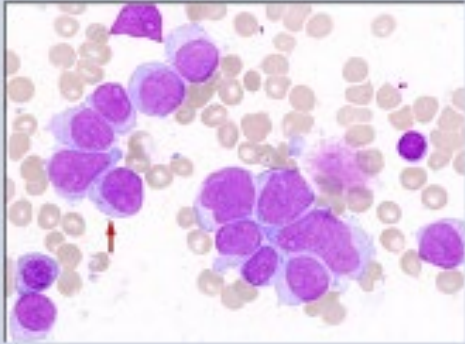
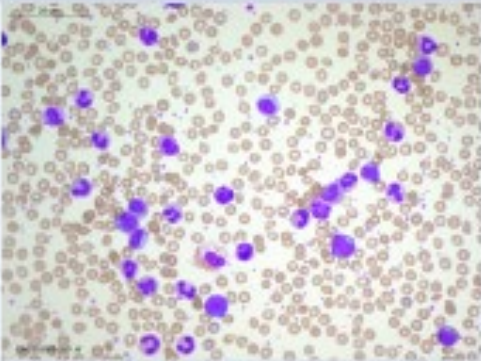
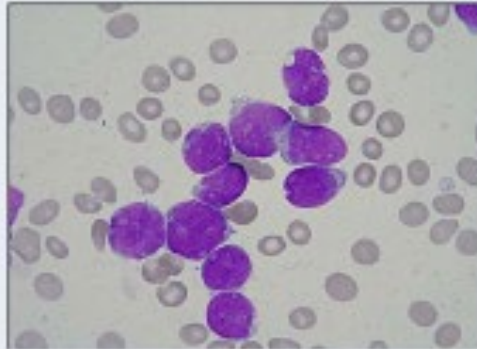
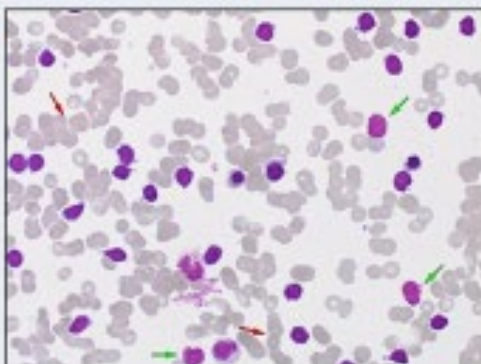


HIGHLY
COMMENDED 2020

It's a Mocking Bird
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What is Leukaemia?

- Leukaemia is a **clonal** malignant disease of the haematopoietic system
- May arise from the **lymphoid** or **myeloid** lineage

	Acute	Chronic
Myeloid		
Lymphoid		

Clinical features of Acute leukaemia

1. Bone marrow failure	Anaemia Infection Bleeding (DIC in acute promyelocytic leukaemia)
2. Extra-medullary infiltration to other organs	liver, spleen, lymph nodes, skin, gum



- **AML**
 - Gum hypertrophy (acute monoblastic leukaemia)
- **ALL**
 - Hepatosplenomegaly, lymphadenopathy
 - Mediastinal mass (T-ALL)
 - CNS, testicular involvement

Workup for suspected acute leukaemia

- 1) Make **Diagnosis**
- 2) Watch out for **Haematological Emergencies**
- 3) **Prepare** patient for **treatment**

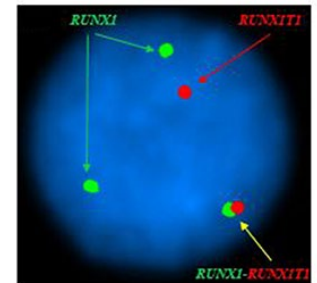
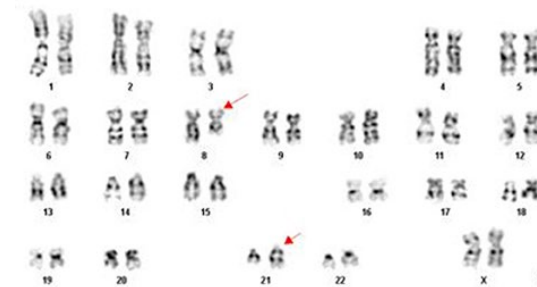
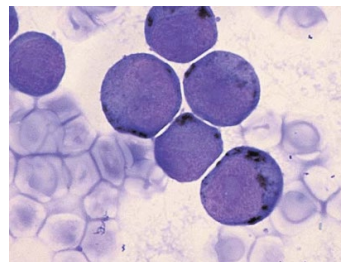
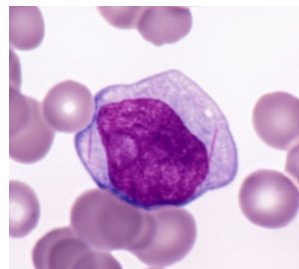
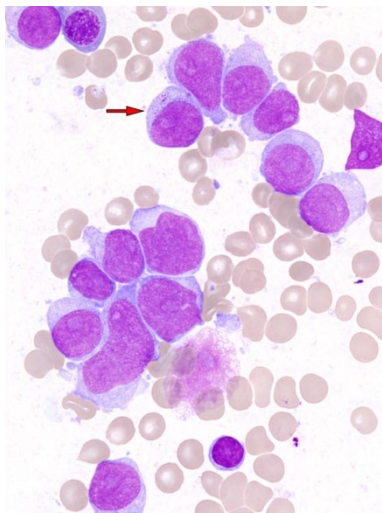
- CBP + differential (WBC high/normal/low) + manual count
- Clotting profile, d-dimer, fibrinogen (DIC in APL)
- Biochemistry – renal function, potassium, calcium, phosphate, urate, LDH (features of tumour lysis syndrome)
- Bone marrow examination + cytogenetics + molecular/NGS

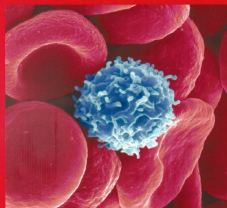
- CXR (disease related complications, infections)
- ECG, echocardiogram (before anthracycline)
- Hepatitis serology, HIV
- G6PD (risk of oxidative hemolysis with co-trimoxazole)
- Arrange central venous catheter insertion
- HLA typing for patients with high risk disease and candidates for HSCT

Workup for suspected acute leukaemia

Bone Marrow Examination

- **Morphology** (APL, AML, ALL) on PB and BM
- **Cytochemistry** (MPO and SBB) – AML vs ALL
- **Flow cytometry** – for sub-classification lineage specification
- **Cytogenetics** (and FISH to detect specific chromosomal abnormalities) – diagnostic and prognostic information
- **Molecular genetics** – diagnostic and prognostic information





WHO Classification of Haematolymphoid Tumours,
5th edition, 2022

SPRINGER NATURE

WHO Classification of Haematolymphoid Tumours, 5th edition, 2022

Leukemia

www.nature.com/leu

REVIEW ARTICLE OPEN

The 5th edition of the World Health Organization Classification of Haematolymphoid Tumours: Myeloid and Histiocytic/Dendritic Neoplasms

Joseph D. Khoury¹, Eric Solary², Oussama Abal³, Yasmine Akkari⁴, Rita Alaggio⁵, Jane F. Apperley⁶, Rafael Bejar⁷, Emilio Berti⁸, Lambert Busque⁹, John K. C. Chan¹⁰, Weina Chen¹¹, Xueyan Chen¹², Wee-Joo Chng¹³, John K. Choi¹⁴, Isabel Colmenero¹⁵, Sarah E. Coupland¹⁶, Nicholas C. P. Cross¹⁷, Daphne De Jong¹⁸, M. Tarek Elghetany¹⁹, Emilio Takahashi²⁰, Jean-François Emile²¹, Judith Ferry²², Linda Fogelstrand²³, Michaela Fontenay²⁴, Ulrich Germing²⁵, Sumeet Gujral²⁶, Torsten Haferlach²⁷, Claire Harrison²⁸, Jennelle C. Hodge²⁹, Shimin Hu³⁰, Joop H. Jansen³¹, Rashmi Kanagal-Shamanna³², Hagop M. Kantarjian³³, Christian P. Kratz³⁴, Xiao-Qiu Li³⁵, Megan S. Lim³⁶, Keith Loeb³⁷, Sanam Loghavi³⁸, Andrea Marcogliese³⁹, Soheil Meshkini⁴⁰, Phillip Michaels⁴¹, Kikkeri N. Nares⁴², Yasodha Natkunam⁴³, Reza Nejjari⁴⁴, German Ott⁴⁵, Eric Padron⁴⁶, Keyur P. Patel⁴⁷, Nikhil Parkar⁴⁸, Jennifer Picarsic⁴⁹, Uwe Platzbecker⁵⁰, Irene Roberts⁵¹, Anna Schuh⁵², William Sewell⁵³, Reiner Siebert⁵⁴, Prashant Tembhare⁵⁵, Jeffrey Tyner⁵⁶, Srdan Verstovsek⁵⁷, Wei Wang⁵⁸, Brent Wood⁵⁹, Wenbin Xiao⁶⁰, Cecilia Yeung⁶¹ and Andreas Hochhaus⁶²

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Table 8. Cytogenetic and molecular abnormalities defining acute myeloid leukaemia, myelodysplasia-related.

Defining cytogenetic abnormalities
Complex karyotype (≥3 abnormalities)
5q deletion or loss of 5q due to unbalanced translocation
Monosomy 7, 7q deletion, or loss of 7q due to unbalanced translocation
11q deletion
12p deletion or loss of 12p due to unbalanced translocation
Monosomy 13 or 13q deletion
17p deletion or loss of 17p due to unbalanced translocation
Isochromosome 17q
idic(X)(q13)
Defining somatic mutations
ASXL1
BCOR
EZH2
SF3B1
SRSF2
STAG2
U2AF1
ZRSR2

Table 7. Acute myeloid leukaemia.

Acute myeloid leukaemia with defining genetic abnormalities
Acute promyelocytic leukaemia with <i>PML::RARA</i> fusion
Acute myeloid leukaemia with <i>RUNX1::RUNX1T1</i> fusion
Acute myeloid leukaemia with <i>CBFB::MYH11</i> fusion
Acute myeloid leukaemia with <i>DEK::NUP214</i> fusion
Acute myeloid leukaemia with <i>RBM15::MRTFA</i> fusion
Acute myeloid leukaemia with <i>BCR::ABL1</i> fusion
Acute myeloid leukaemia with <i>KMT2A</i> rearrangement
Acute myeloid leukaemia with <i>MECOM</i> rearrangement
Acute myeloid leukaemia with <i>NUP98</i> rearrangement
Acute myeloid leukaemia with <i>NPM1</i> mutation
Acute myeloid leukaemia with <i>CEBPA</i> mutation
Acute myeloid leukaemia, myelodysplasia-related
Acute myeloid leukaemia with other defined genetic alterations
Acute myeloid leukaemia, defined by differentiation
Acute myeloid leukaemia with minimal differentiation
Acute myeloid leukaemia without maturation
Acute myeloid leukaemia with maturation
Acute basophilic leukaemia
Acute myelomonocytic leukaemia
Acute monocytic leukaemia
Acute erythroid leukaemia
Acute megakaryoblastic leukaemia

Table 10. Subtypes of myeloid neoplasms associated with germline predisposition.

Myeloid neoplasms with germline predisposition without a pre-existing platelet disorder or organ dysfunction
• Germline <i>CEBPA</i> P/LP variant (CEBPA-associated familial AML)
• Germline <i>DDX41</i> P/LP variant ^a
• Germline <i>TP53</i> P/LP variant ^a (Li-Fraumeni syndrome)
Myeloid neoplasms with germline predisposition and pre-existing platelet disorder
• Germline <i>RUNX1</i> P/LP variant ^a (familial platelet disorder with associated myeloid malignancy, FPD-MM)
• Germline <i>ANKRD26</i> P/LP variant ^a (Thrombocytopenia 2)
• Germline <i>ETV6</i> P/LP variant ^a (Thrombocytopenia 5)
Myeloid neoplasms with germline predisposition and potential organ dysfunction
• Germline <i>GATA2</i> P/LP variant (GATA2-deficiency)
• Bone marrow failure syndromes
◦ Severe congenital neutropenia (SCN)
◦ Shwachman-Diamond syndrome (SDS)
◦ Fanconi anaemia (FA)
• Telomere biology disorders
• RASopathies (Neurofibromatosis type 1, CBL syndrome, Noonan syndrome or Noonan syndrome-like disorders ¹⁵)
• Down syndrome ¹⁶
• Germline <i>SAMD9</i> P/LP variant (MIRAGE Syndrome)
• Germline <i>SAMD9L</i> P/LP variant (SAMD9L-related Ataxia Pancytopenia Syndrome) ¹⁷
• Biallelic germline <i>BLM</i> P/LP variant (Bloom syndrome)

Leukemia

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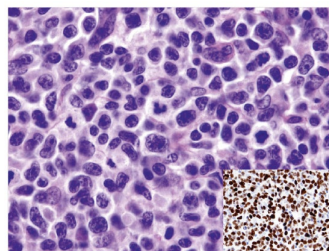
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LYMPHOMA

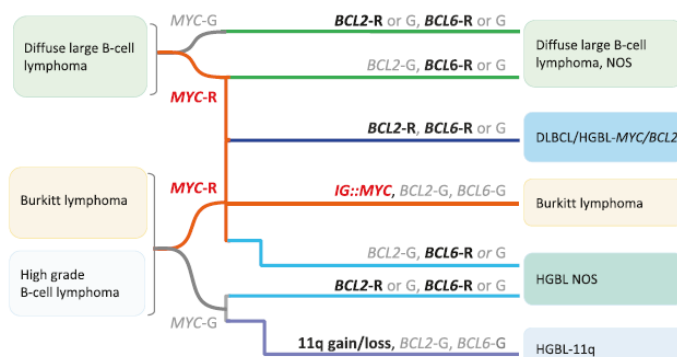
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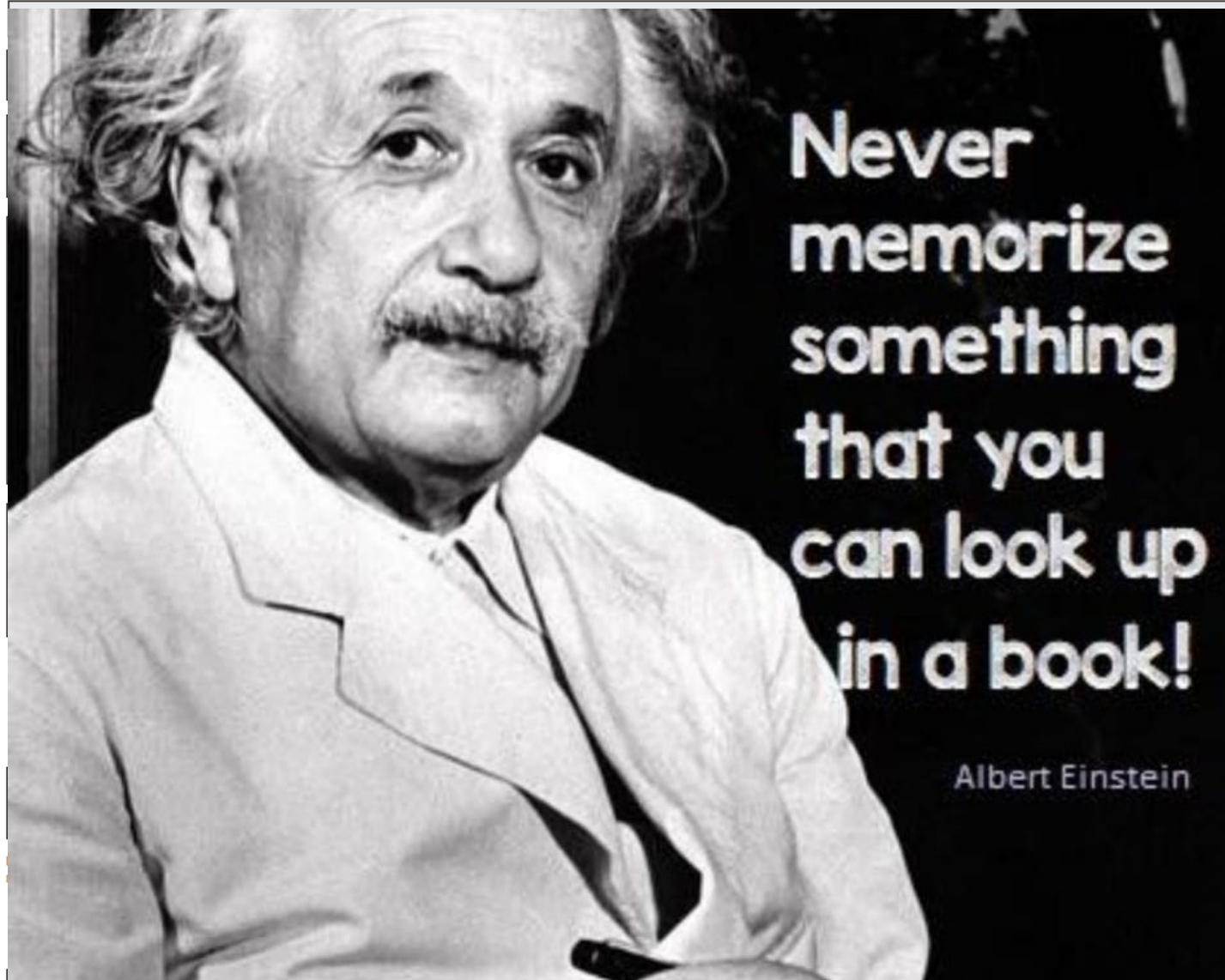
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Morphology → Genetic changes → Diagnosis



Approach to the classification of acute myeloid leukaemia



Never
memorize
something
that you
can look up
in a book!

Albert Einstein

Step 2: Diagnostic qualifier appended to any of the diagnoses

Post cytotoxic therapy or therapy-related

Post-MDS or MDS/MPN

Associated with germline gene variant or genetic predisposition

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graph TD; A[Treatment for acute leukaemia] --> B[Supportive]; A --> C[Specific]
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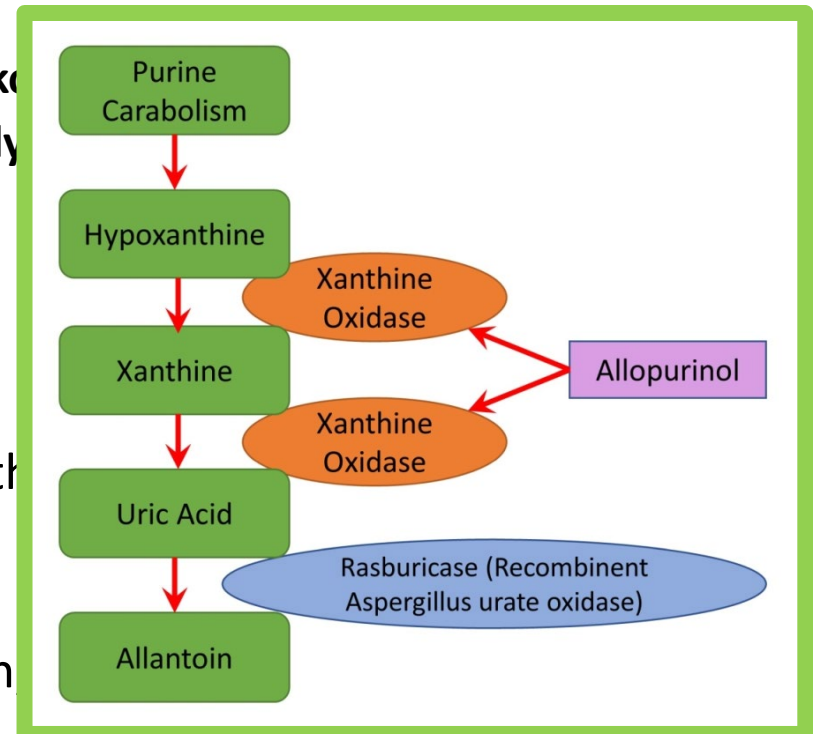
Treatment for
acute
leukaemia

Supportive

Specific

Supportive treatment

- Watch out for **haematological emergencies**
 1. Resuscitation if in **septic shock**
 2. Immediate empirical broad spectrum antibiotics for **neutropenic sepsis**
 3. Hydration and urate lowering agents (allopurinol, febuxostat, rasburicase) to prevent **tumour lysis syndrome**
 4. Urgent leukapheresis if signs of **leukaemia**
 5. **Disseminated intravascular haemolytic leukaemia (APL)**
- RBC and platelet **transfusions**
- **Antifungal prophylaxis** in patients with neutropenia
- **Proper nursing care** - reverse isolation, bacteria diet



CLINICAL CASE

Mr. A

HAEMATOLOGY REPORT

Arrive Date :	15/09/15	29/05/18		
Arrive Time :	15:20	19:05		
Request No. :	HY579169	HY344024	Reference	Units
Urgency :	--	URGENT	Interval	
Specimen type: EDTA blood				
COMPLETE BLOOD COUNT				
WBC	6.6	86.5 H	3.7 - 9.2	x10 ⁹ /L
RBC	5.32	3.78	4.30 - 5.90	x10 ¹² /L
HGB	14.6	11.0 L	13.4 - 17.1	g/dL
HCT	0.428	0.309 L	0.400 - 0.510	L/L
MCV	80.5 L	81.7 L	82.0 - 97.0	fL
MCH	27.4	29.1	27.0 - 33.0	pg
MCHC	34.1	35.6 H	32.0 - 35.0	g/dL
RDW	14.1 H	17.9 H	11.0 - 14.0	%
PLT	235	27 L	145 - 370	x10 ⁹ /L
WBC DIFFERENTIAL				
Neutrophil	4.3	0.9 L	1.7 - 5.8	x10 ⁹ /L
NEU %	64.3	1.0	43.0 - 71.0	%
Lymphocyte	1.9	6.9 H	1.0 - 3.1	x10 ⁹ /L
LYM %	29.0	8.0 L	19.0 - 46.0	%
Monocyte	0.3		0.1 - 0.8	x10 ⁹ /L
MON %	4.4		3.0 - 12.0	%
Eosinophil	0.1		0.0 - 0.5	x10 ⁹ /L
EOS %	1.5		0 - 7.0	%
Basophil	0.1		0.0 - 0.1	x10 ⁹ /L
BAS %	0.8		0 - 1.4	%
Blast		78.72		x10 ⁹ /L
BLAST %		91.0		%
Remarks	N			
Comment:				
18HY344024 Differential count no result. Suggest repeat CBC in normal hours.				
Blasts predominating, mainly medium sized, fine chromatin, round to mild nuclear contour, few small nucleoli. No Auer rod or granules seen. Some smear cells observed.				
Thrombocytopenia confirmed				

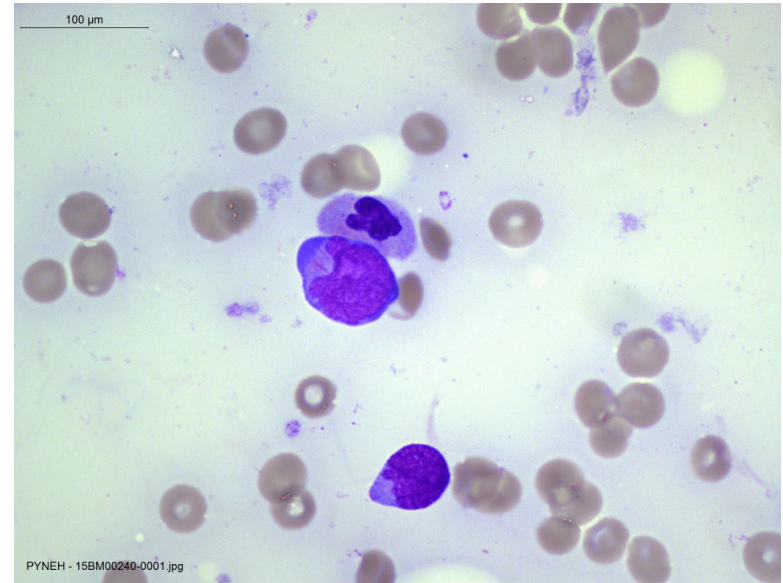
HAEMATOLOGY REPORT

- M/58
- Good past health
- Presented with unresolved fever and upper respiratory tract infection for 2 weeks
- Physical examination showed lower limb petechiae, tachycardia and hypotensive
- What is the abnormality ?
 - Leukocytosis
 - Thrombocytopenia
 - Neutropenia
 - Circulating blasts 90%
- What investigation will you order to confirm the diagnosis ?
 - Bone marrow examination

Mr. A

Arrive Date :	15/09/15	29/05/18		
Arrive Time :	15:20	19:05		
Request No. :	HY579169	HY344024	Reference	
Urgency :	--	URGENT	Interval	Units
Specimen type: EDTA blood				
COMPLETE BLOOD COUNT				
WBC	6.6	86.5 H	3.7 - 9.2	x10 ⁹ /L
RBC	5.32	3.78 L	4.30 - 5.90	x10 ¹² /L
HGB	14.6	11.0 L	13.4 - 17.1	g/dL
HCT	0.428	0.309 L	0.400 - 0.510	L/L
MCV	80.5 L	81.7 L	82.0 - 97.0	fL
MCH	27.4	29.1	27.0 - 33.0	pg
MCHC	34.1	35.6 H	32.0 - 35.0	g/dL
RDW	14.1 H	17.3 H	11.0 - 14.0	%
PLT	235	27 L	145 - 370	x10 ⁹ /L
WBC DIFFERENTIAL				
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NEU %	64.3	1.0 L	43.0 - 71.0	%
Lymphocyte	1.9	6.9 H	1.0 - 3.1	x10 ⁹ /L
LYM %	29.0	8.0 L	19.0 - 46.0	%
Monocyte	0.3		0.1 - 0.8	x10 ⁹ /L
MON %	4.4		3.0 - 12.0	%
Eosinophil	0.1		0.0 - 0.5	x10 ⁹ /L
EOS %	1.5		0 - 7.0	%
Basophil	0.1		0.0 - 0.1	x10 ⁹ /L
BAS %	0.8		0 - 1.4	%
Blast		78.72		x10 ⁹ /L
BLAST %		91.0		%
Remarks	N	Y		
Comment:				
18HY344024 Differential count no result. Suggest repeat CBC in normal hours.				
Blasts predominating, mainly medium sized, fine chromatin, round to mild nuclear contour, few small nucleoli. No Auer rod or granules seen. Some smear cells observed. Thrombocytopenia confirmed				

HAEMATOLOGY REPORT



The diagnosis is **acute myeloid leukaemia**.
Complicated by **neutropenic fever** and
impending septic shock.

Haematological Emergencies1

Neutropenic fever

- One of the commonest haemato-oncological emergencies
- Defined as fever in a patient with an absolute neutrophil count (ANC) $<1 \times 10^9/L$
- Can be due to both **gram positive** (esp. in patients with central catheters) and **gram negative** organism
- Need immediate medical attention
- Septic workup and start **IV antibiotics ASAP**
- Fluid resuscitation and/or inotropes



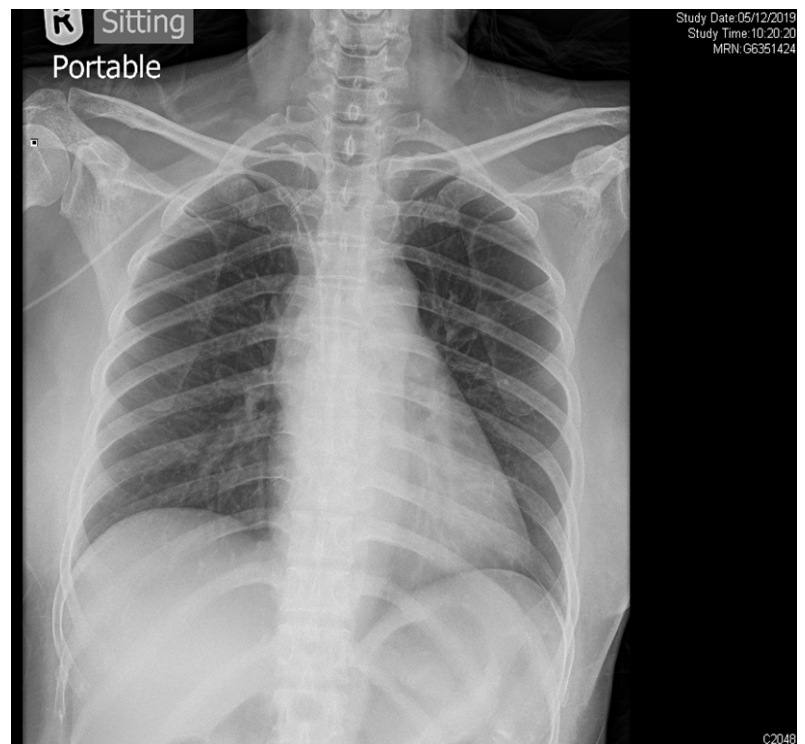
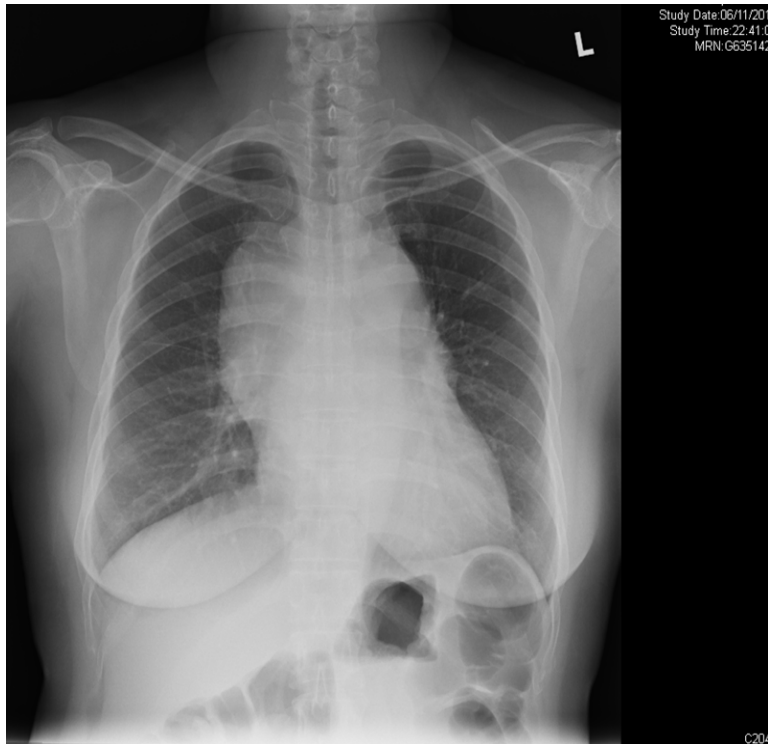
Ms. B

- F/28
- Good past health
- Presented with unresolved fever and shortness of breath
- Physical examination showed cervical lymphadenopathy, splenomegaly 2cm below costal margin
- What are the abnormalities ?
 - **Marked leucocytosis**
 - **Anaemia**
 - **Thrombocytopenia**
 - **Circulating blasts**
- What investigation will you order to confirm the diagnosis ?
 - **Bone marrow examination**

Arrive Date :	24/02/07	29/01/09	06/04/10	27/01/12	03/10/19		
Arrive Time :	10:34	10:03	11:21	11:42	11:27		
Request No. :	HR006924	HR003761	HR808661	HR003364	HR064498	Reference	Units
Urgency :	--	URGENT	URGENT	--	URGENT	Interval	
CBC (Specimen: Blood)							
WBC	7.0	6.5	7.9	6.4	540.95 C	3.7 - 9.3	10 ⁹ /L
WBC						3.7-9.3	x10 ⁹ /L
RBC	4.34	4.11	4.58	4.42	2.54 L	3.9 - 5.2	10 ¹² /L
RBC						3.90-5.20	x10 ¹² /L
HGB	13.7	13.3	14.1	13.8	6.5 L	11.5-15.4	g/dL
HCT	0.396	0.371	0.418	0.398	0.236 L	0.34-0.46	L/L
MCV	91.4	90.2	91.3	90.0	92.9	80.0-96.0	fL
MCH	31.7	32.3	30.8	31.2	25.6 L	26.0-33.0	pg
MCHC	34.7	35.8	33.7	34.7	27.5 L	32.0-35.6	g/dL
RDW	12.2	12.7	12.1	12.0	17.2 H	11.8-15.8	%
PLT	303	283	261	286	62 L	160 - 420	10 ⁹ /L
PLT						160-420	x10 ⁹ /L
MPV	7.7	7.3	9.7	9.9	9.3	8.1-11.5	fL
WBC DIFFERENTIAL							
Neutrophil	4.2	3.9	5.1	3.86		1.8 - 6.2	10 ⁹ /L
Neutrophil					30.65 H	1.8-6.2	x10 ⁹ /L
NEU%	60.1	60.5	64.1			44.0 - 72.0	%
Lymphocyte	2.1	1.9	2.3	2.04		1.0 - 3.2	10 ⁹ /L
Lymphocyte					84.75 H	1.0-3.2	x10 ⁹ /L
LYM%	30.2	29.8	28.8			20.0 - 45.0	%
Monocyte	0.5	0.4	0.4	0.31		0.2 - 0.7	10 ⁹ /L
MONO%	6.6	6.3	4.9			2.0 - 10.0	%
Eosinophil	0.2	0.2	0.2	0.17		0.0 - 0.6	10 ⁹ /L
EOS%	2.6	2.7	1.9			1.0 - 6.0	%
Basophil	0.0	0.0	0.0	0.02		0.0 - 0.1	10 ⁹ /L
BASO%	0.5	0.7	0.3			< 2	%
Blast					423.74 H		x10 ⁹ /L
Metamyelocyte					1.80 H		x10 ⁹ /L
DC Type	MACHINE	MACHINE	MACHINE	MACHINE	MANUAL		
NRBC/100WBC					1		
Film Review	N	N	N	N	Y		
Comment:							
1049064498 Report Comment							

HAEMATOLOGY REPORT

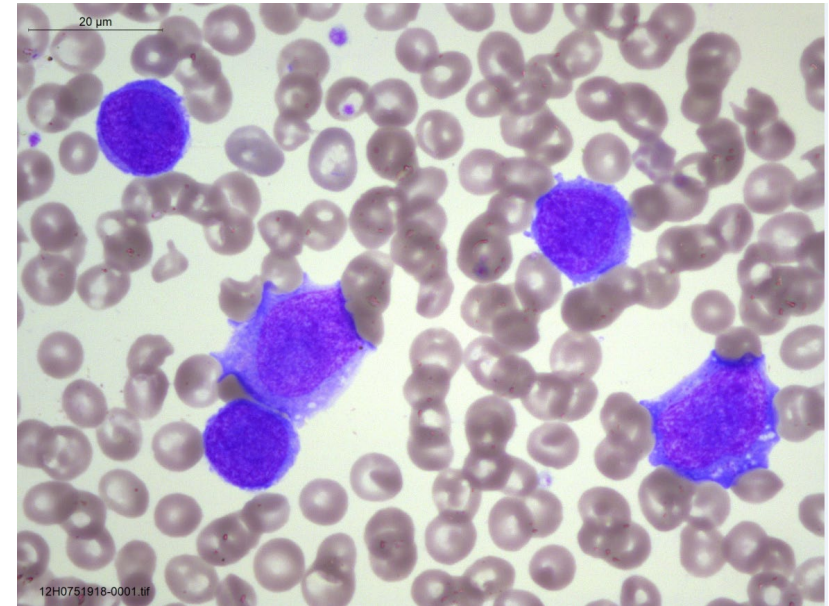
Ms. B



Ms. B

Arrive Date :	24/02/07	29/01/09	06/04/10	27/01/12	03/10/19		
Arrive Time :	10:34	10:03	11:21	11:42	11:27		
Request No. :	HR006924	HR003761	HR808661	HR003364	HR064498	Reference	Units
Urgency :	--	URGENT	URGENT	--	URGENT	Interval	
CBC (Specimen: Blood)							
WBC	7.0	6.5	7.9	6.4		3.7 - 9.3	10 ⁹ /L
WBC					540.95 C	3.7-9.3	x10 ⁹ /L
RBC	4.34	4.11	4.58	4.42		3.9 - 5.2	10 ¹² /L
RBC					2.54 L	3.90-5.20	x10 ¹² /L
HGB	13.7	13.3	14.1	13.8	6.5 L	11.5-15.4	g/dL
HCT	0.396	0.371	0.418	0.398	0.236 L	0.34-0.46	L/L
MCV	91.4	90.2	91.3	90.0	92.9	80.0-96.0	fL
MCH	31.7	32.3	30.8	31.2	25.6 L	26.0-33.0	pg
MCHC	34.7	35.8	33.7	34.7	27.5 L	32.0-35.6	g/dL
RDW	12.2	12.7	12.1	12.0	17.2 H	11.8-15.8	%
PLT	303	283	261	286		160 - 420	10 ⁹ /L
PLT					62 L	160-420	x10 ⁹ /L
MPV	7.7	7.3	9.7	9.9	9.3	8.1-11.5	fL
WBC DIFFERENTIAL							
Neutrophil	4.2	3.9	5.1	3.86		1.8 - 6.2	10 ⁹ /L
Neutrophil					30.65 H	1.8-6.2	x10 ⁹ /L
NEU%	60.1	60.5	64.1			44.0 - 72.0	%
Lymphocyte	2.1	1.9	2.3	2.04		1.0 - 3.2	10 ⁹ /L
Lymphocyte					84.75 H	1.0-3.2	x10 ⁹ /L
LYM%	30.2	29.8	28.8			20.0 - 45.0	%
Monocyte	0.5	0.4	0.4	0.31		0.2 - 0.7	10 ⁹ /L
MONO%	6.6	6.3	4.9			2.0 - 10.0	%
Eosinophil	0.2	0.2	0.2	0.17		0.0 - 0.6	10 ⁹ /L
EOS%	2.6	2.7	1.9			1.0 - 6.0	%
Basophil	0.0	0.0	0.0	0.02		0.0 - 0.1	10 ⁹ /L
BASO%	0.5	0.7	0.3			< 2	%
Blast					423.74 H		x10 ⁹ /L
Metamyelocyte					1.80 H		x10 ⁹ /L
DC Type	MACHINE	MACHINE	MACHINE	MACHINE	MANUAL		
NRBC/100WBC					1		
Film Review	N	N	N	N	Y		
Comment:							
10UB064498 Report Comment							

HAEMATOLOGY REPORT



The diagnosis is
T-Acute Lymphoblastic Leukaemia
with marked leukocytosis.

Ms. B

Collect Date :	25/10/19	25/10/19		
Collect Time :	22:04	22:04		
Request No. :	CA260512	CA261104		
Remark :	AKI.Mask	AKI	Ref. Interval	Units
	Type =			
	Room			
	Air,Flo...			
Comment	Below			
Mask Type	Room Air			
Flow Rate	----			L/min
% inspired O2	21			%
pH	7.38	7.35 - 7.45		
pO2	3.7 L	10.6 - 14.0		kPa
pCO2	5.5	4.7 - 6.0		kPa
HCO3-	24	22 - 26		mmol/L
Base excess	-1	(-4) - (+2)		mmol/L
Na	136	136 - 148		mmol/L
K	5.5 H	3.6 - 5.0		mmol/L
Chloride	100	100 - 109		mmol/L
Urea	22.4 H	2.5 - 6.4		mmol/L
Creatinine	369 H	49 - 82		umol/L
Estimated GFR	13 L	>90		unit
Calcium	2.11	2.11 - 2.55		mmol/L
Adjusted Calcium	2.43	2.11 - 2.55		mmol/L
Phosphate	1.73 H	0.88 - 1.45		mmol/L
Total Protein	59 L	67 - 87		g/L
Albumin	26 L	39 - 50		g/L
Globulin	33	26 - 40		g/L
Total Bilirubin	15	4 - 23		umol/L
ALP	136 H	32 - 93		U/L
ALT	15	7 - 36		U/L
AST	34 H	14 - 30		U/L
LDH	729 H	107 - 218		U/L
Urate	1161 H	177 - 400		umol/L
Comment:	19CA260512 ? Venous Blood. Reference range is for arterial blood only. Pls send only ARTERIAL BLOOD.			
Footnotes:	5555 Comment:			

What are the abnormalities?

- Hyperkalaemia,
- Acute renal failure,
- Hyperphosphataemia,
- Hyperuricaemia,
- Raised LDH

What is the diagnosis?

- Tumour Lysis syndrome

Haematological Emergencies 2

Tumour lysis syndrome

- A set of **metabolic complications** resulting from a rapidly proliferating and drug-sensitive neoplasm
- Can also be spontaneous
- **Hyperuricaemia, hyperkalaemia, hyperphosphataemia, hypocalcaemia, raised LDH**
- **Acute renal failure, arrhythmia, seizure, confusion**
- **Prevention & Treatment**
 - Aggressive hydration
 - Allopurinol , febuxostat (xanthine oxidase inhibitor)
 - Rasburicase (recombinant urate oxidase, c/i in G6PD deficiency)
 - Renal replacement therapy (haemodialysis)
- **Monitor** WBC or tumour size, fluid status, urine output, electrolytes, RFT, LDH, pH

Ms. B

Arrive Date :	24/02/07	29/01/09	06/04/10	27/01/12	03/10/19		
Arrive Time :	10:34	10:03	11:21	11:42	11:27		
Request No. :	HR006924	HR003761	HR808661	HR003364	HR064498	Reference	Units
Urgency :	--	URGENT	URGENT	--	URGENT	Interval	
CBC (Specimen: Blood)							
WBC	7.0	6.5	7.9	6.4	540.95 C	3.7 - 9.3	10 ⁹ /L
RBC	4.34	4.11	4.58	4.42	2.54 L	3.7-9.3	x10 ⁹ /L
HGB	13.7	13.3	14.1	13.8	6.5 L	3.9 - 5.2	10 ¹² /L
HCT	0.396	0.371	0.418	0.398	0.236 L	3.90-5.20	x10 ¹² /L
MCV	91.4	90.2	91.3	90.0	92.9	11.5-15.4	g/dL
MCH	31.7	32.3	30.8	31.2	25.6 L	0.34-0.46	L/L
MCHC	34.7	35.8	33.7	34.7	27.5 L	80.0-96.0	fL
RDW	12.2	12.7	12.1	12.0	17.2 H	26.0-33.0	pg
PLT	303	283	261	286	62 L	32.0-35.6	g/dL
MPV	7.7	7.3	9.7	9.9	9.3	11.8-15.8	%
WBC DIFFERENTIAL							
Neutrophil	4.2	3.9	5.1	3.86	30.65 H	1.8 - 6.2	10 ⁹ /L
Neutrophil						1.8-6.2	x10 ⁹ /L
NEU%	60.1	60.5	64.1			44.0 - 72.0	%
Lymphocyte	2.1	1.9	2.3	2.04		1.0 - 3.2	10 ⁹ /L
Lymphocyte					84.75 H	1.0-3.2	x10 ⁹ /L
LYM%	30.2	29.8	28.8			20.0 - 45.0	%
Monocyte	0.5	0.4	0.4	0.31		0.2 - 0.7	10 ⁹ /L
MONO%	6.6	6.3	4.9			2.0 - 10.0	%
Eosinophil	0.2	0.2	0.2	0.17		0.0 - 0.6	10 ⁹ /L
EOS%	2.6	2.7	1.9			1.0 - 6.0	%
Basophil	0.0	0.0	0.0	0.02		0.0 - 0.1	10 ⁹ /L
BASO%	0.5	0.7	0.3			< 2	%
Blast					423.74 H		x10 ⁹ /L
Metamyelocyte					1.80 H		x10 ⁹ /L
DC Type	MACHINE	MACHINE	MACHINE	MACHINE	MANUAL		
NRBC/100WBC					1		
Film Review	N	N	N	N	Y		
Comment:							
10/07/2019 Report Comment							

HAEMATOLOGY REPORT

- Developed **desaturation and mental dullness** after admission
- CXR showed diffuse lung infiltrates
- CT brain showed no haemorrhage
- What is the diagnosis?
Leukostasis (hyperleukostasis)
- How would you manage?
 - Leukapheresis**
 - Cytoreduction with chemotherapy**
 - TLS prophylaxis**

Haematological Emergencies 3

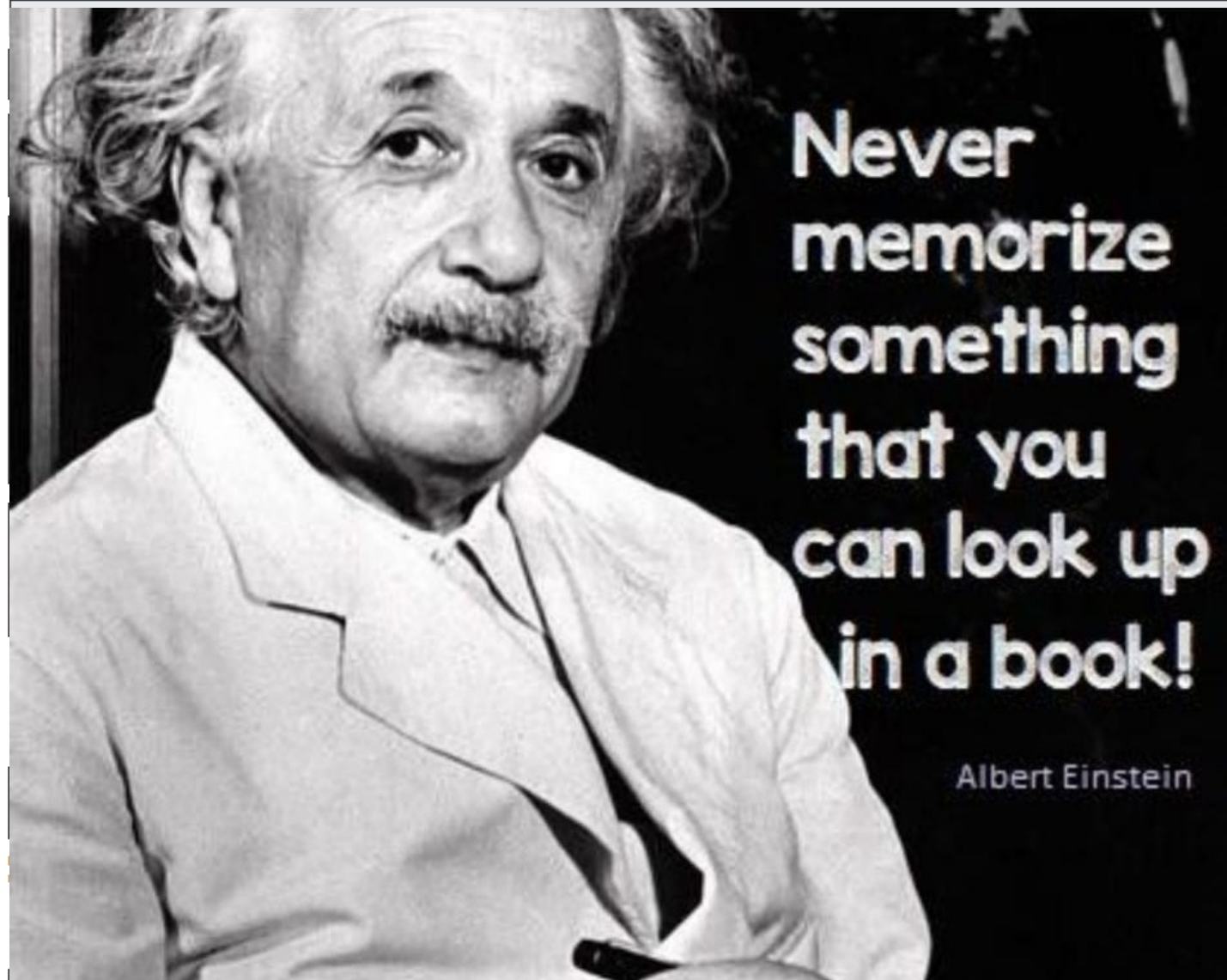
Leukostasis

- Refers to impaired tissue oxygenation **due to sludging of leukocytes in the microcirculation** of a patient with high white blood cell count
- Depends on cell pliability
- **Pulmonary** leukostasis : progressive dyspnoea, hypoxia and diffuse interstitial infiltrates on CXR
- **Cerebral** leukostasis : lethargy, headache, impaired mental state, visual /hearing impairment
- **Central retinal vein thrombosis, priapism**
- **BEWARE:** Anaemia may protect a patient with marked leukostasis from the effect of high whole blood viscosity. Red cell transfusion before cytoreduction may precipitate leukostasis
- **Treatment:** Adequate hydration, urgent leukapheresis and chemotherapy

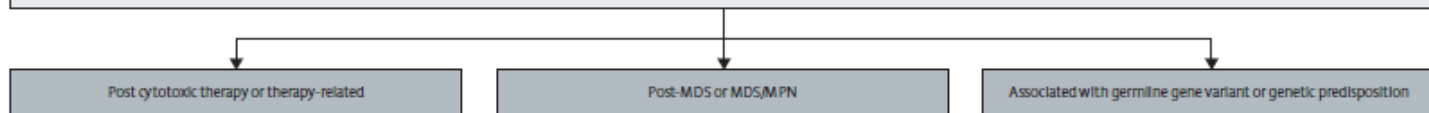
Specific treatment - AML

- **Induction**
 - daunorubicin, cytarabine
 - plus targeted therapy, e.g.
 - midostaurin for FLT3 ITD mutated cases
 - Gemtuzumab ozogamicin for non-transplant favorable risk AML
- **Consolidation**
 - high dose cytarabine
- **Allogeneic HSCT** for high risk cases based on cytogenetics and gene mutation profile, or if relapse

Approach to the classification of acute myeloid leukaemia

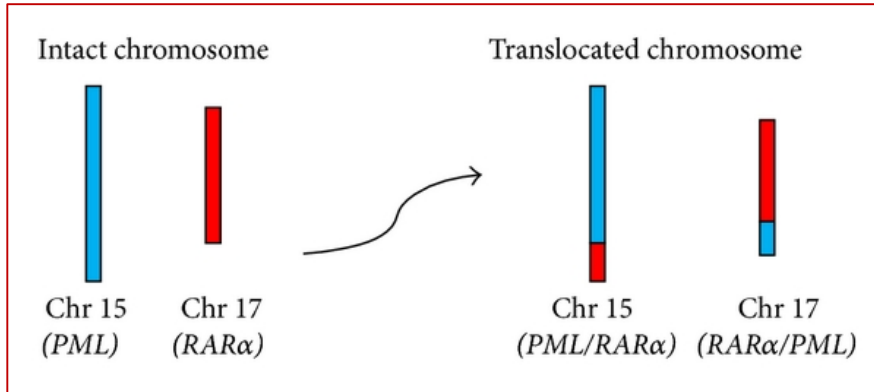


Step 2: Diagnostic qualifier appended to any of the diagnoses



Haematological Emergencies 4

DIC in APL

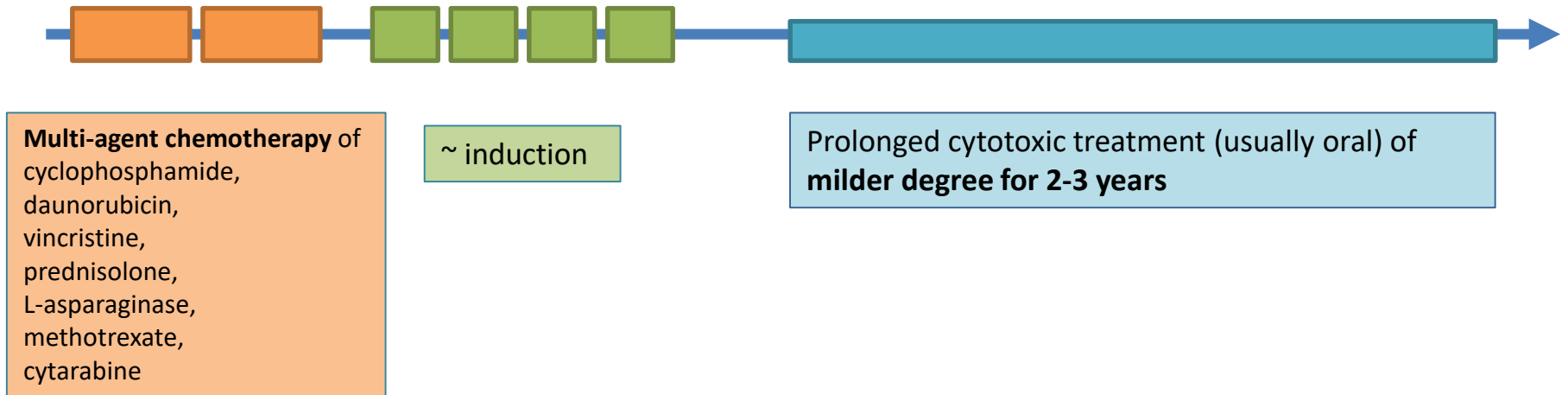


- **Pancytopenia** rather than high WBC
- Prominent bleeding symptoms
- Platelet, PT, APTT, d-dimer, fibrinogen
- Abnormal promyelocytes in circulation

- Characterized by **t(15;17)(q22;q21)**
- The resultant **PML-RARA** fusion protein which blocks RAR-mediated induction of promyelocyte differentiation
- DIC in APL is due to the release procoagulants from the leukaemic blasts
- Requires urgent treatment with **all-trans-retinoic acid (ATRA)** and **arsenic trioxide**
- **Supportive transfusion** with platelet, FFP, cryoprecipitate

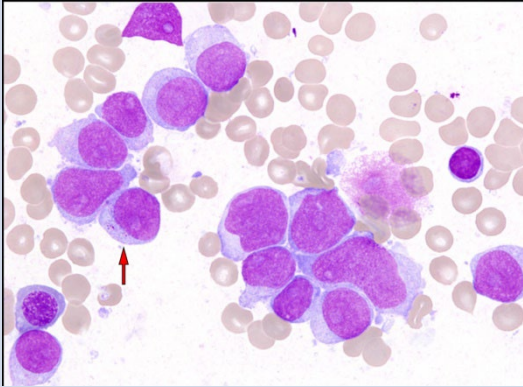
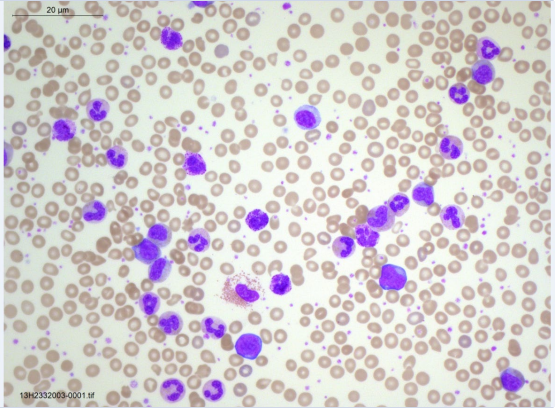
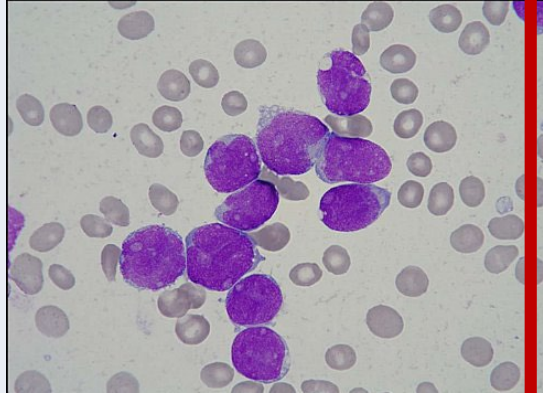
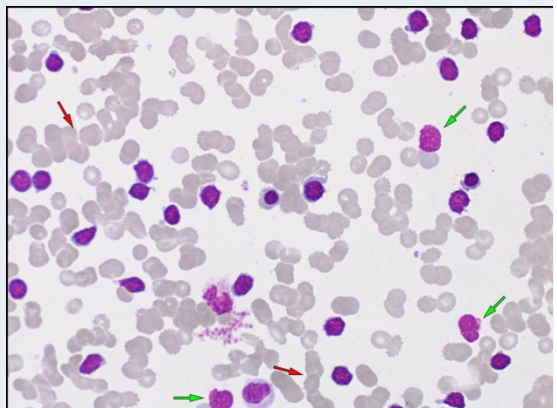
Specific treatment - ALL

Adapted from paediatric protocols



- ❖ **CNS prophylaxis** with intrathecal methotrexate, cranial irradiation, high dose methotrexate (for ALL)
- ❖ **Allogeneic HSCT** for high risk cases, or if relapse

Leukaemia classification

	Acute	Chronic
Myeloid		
Lymphoid		

Ms. C

F/58

Presented with fatigue and weight loss for few months, early satiety and left upper abdominal fullness

Physical examination showed splenomegaly 7cm below costal margin

What are the abnormalities?

- **Marked leukocytosis**
- **Predominant neutrophils and myelocytes**
- **Thrombocytosis**
- **PB Blasts 1%**

Collect Date :	03/07/18		
Collect Time :	22:49		
Request No. :	H1850922	Ref. Interval	Units
CBC			
WBC	254.50 H	3.89 - 9.93	$\times 10^9/L$
RBC	3.34 L	3.87 - 5.11	$\times 10^{12}/L$
HGB	7.9 L	11.5 - 14.8	g/dL
HCT	0.247 L	0.345 - 0.434	
MCV	74.0 L	82.0 - 95.5	fL
MCH	23.6 L	27.0 - 32.4	pg
MCHC	31.8 L	32.1 - 34.9	g/dL
RDW	31.6 H	11.8 - 14.6	%
PLT	608 H	154 - 371	$\times 10^9/L$
WBC DIFFERENTIAL			
DC Type	MANUAL		
#Blast	2.54		$\times 10^9/L$
#Promyelocyte	10.18		$\times 10^9/L$
#Myelocyte	49.63		$\times 10^9/L$
#Metamyelocyte	1.27		$\times 10^9/L$
#Neutrophil	165.43 H	2.01 - 7.42	$\times 10^9/L$
#Lymphocyte	1.27	1.06 - 3.61	$\times 10^9/L$
#Monocyte	2.54 H	0.18 - 0.65	$\times 10^9/L$
#Eosinophil	1.27 H	0.02 - 0.45	$\times 10^9/L$
#Basophil	5.09 H	0.01 - 0.07	$\times 10^9/L$
Total Cell Count	268		
Film Review	Y		
RBC MORPHOLOGY			

HAEMATOLOGY REPORT

Bone Marrow Aspirate:

The marrow is markedly hypercellular for age.

Granulopoiesis is markedly increased, with **bimodal distribution of myelocytes and neutrophils**.

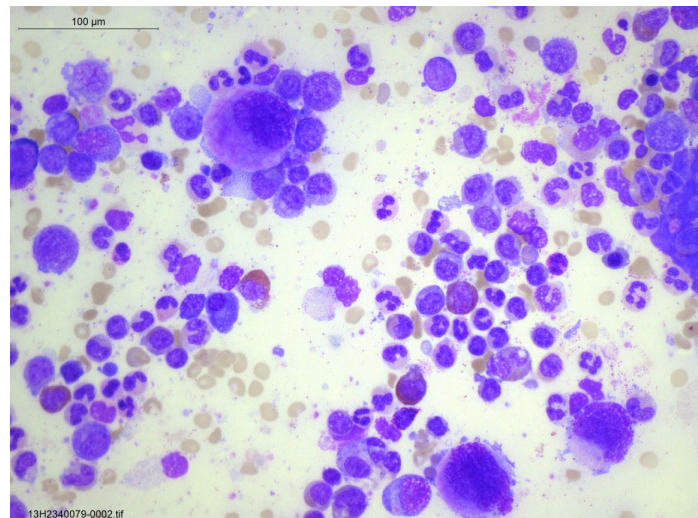
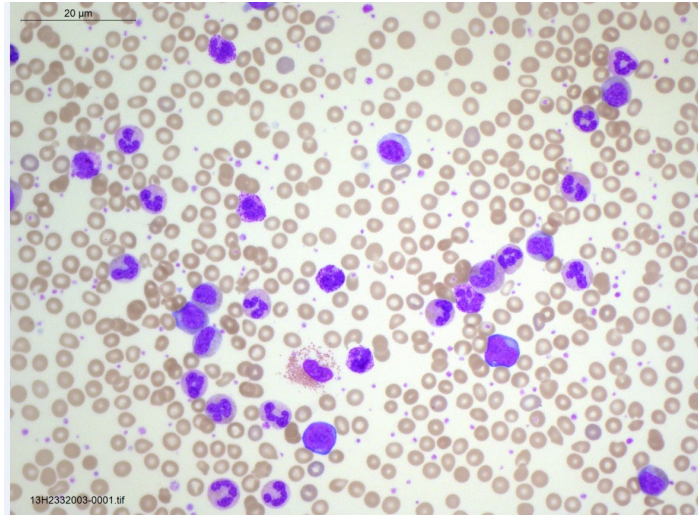
Blasts are not increased (< 1%). Full maturation is attained. **Basophilia** is evident. (2%).

Erythropoiesis is moderately reduced.

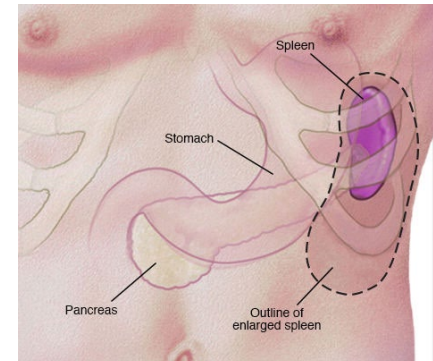
Megakaryopoiesis is markedly increased, with frequent small **hypolobated forms**.

Conclusion: Consistent with **chronic myeloid leukaemia, chronic phase**.

Chronic myeloid leukaemia



- Present with marked **leukocytosis**
 - predominantly neutrophils and myelocytes, basophilia
- **Splenomegaly**



Chronic phase (CP)

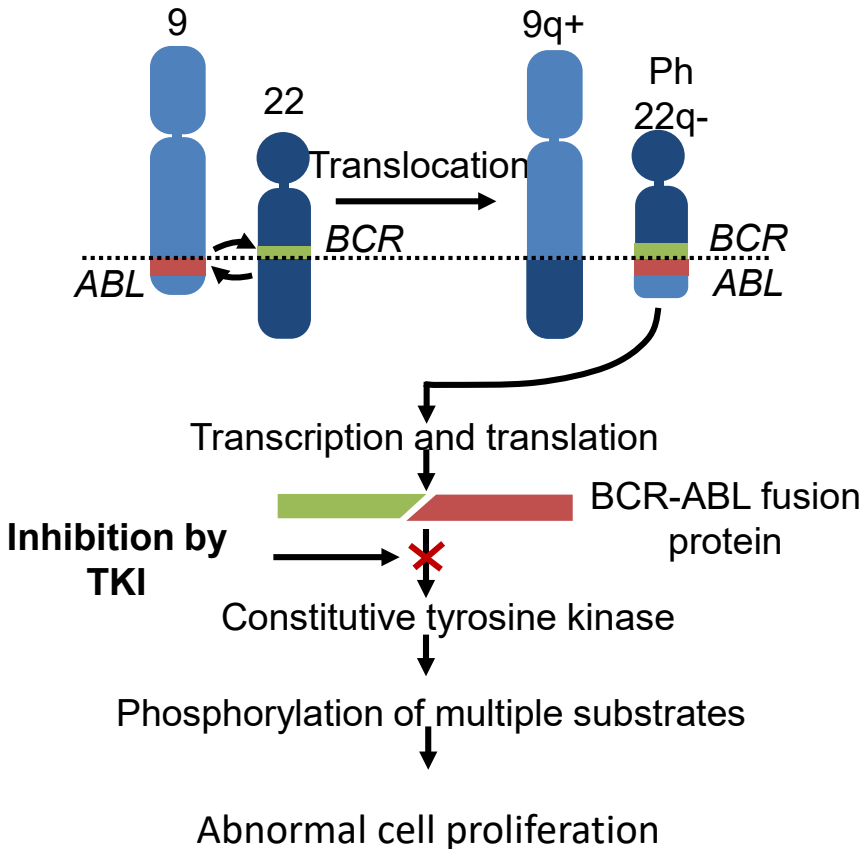


Accelerated phase (AP)



Blastic phase (BC)

Chronic myeloid leukaemia

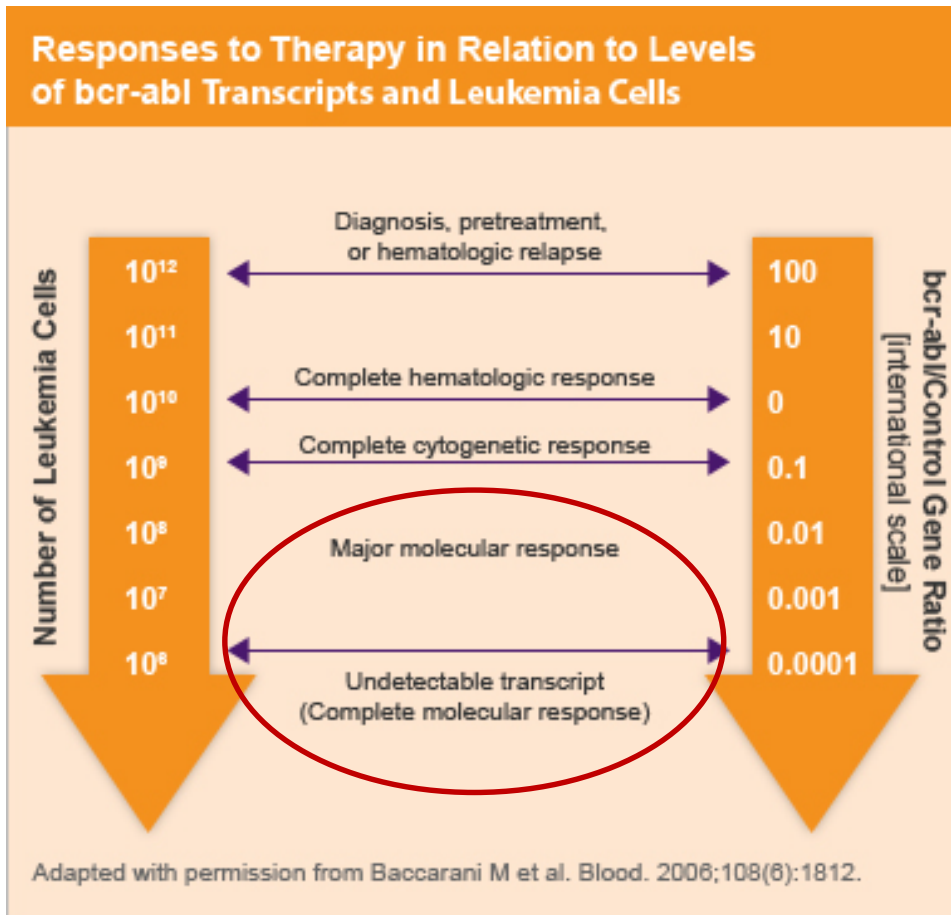


- Defined by **Philadelphia chromosome** **t(9;22)(q34.1;q11.2)**



- The resultant *BCR::ABL* fusion gene product is a **constitutively active tyrosine kinase**
- Specific targeted treatment: tyrosine kinase inhibitor**
 - Imatinib
 - Nilotinib
 - Dasatinib
 - Ponatinib
 - **Asciminib**

Chronic myeloid leukaemia



- **Monitor molecular response** with quantitative reverse transcriptase polymerase chain reaction (RT-PCR)
- Haematopoietic stem cell transplantation is **reserved** for patients not responding to TKI treatment, or in blastic transformation

Mr. D

M/78

Known diabetes mellitus and hypertension

Presented with shortness of breath
on exertion, fatigue and weight
loss for few months

Physical examination showed
cervical and groin

lymphadenopathy, splenomegaly

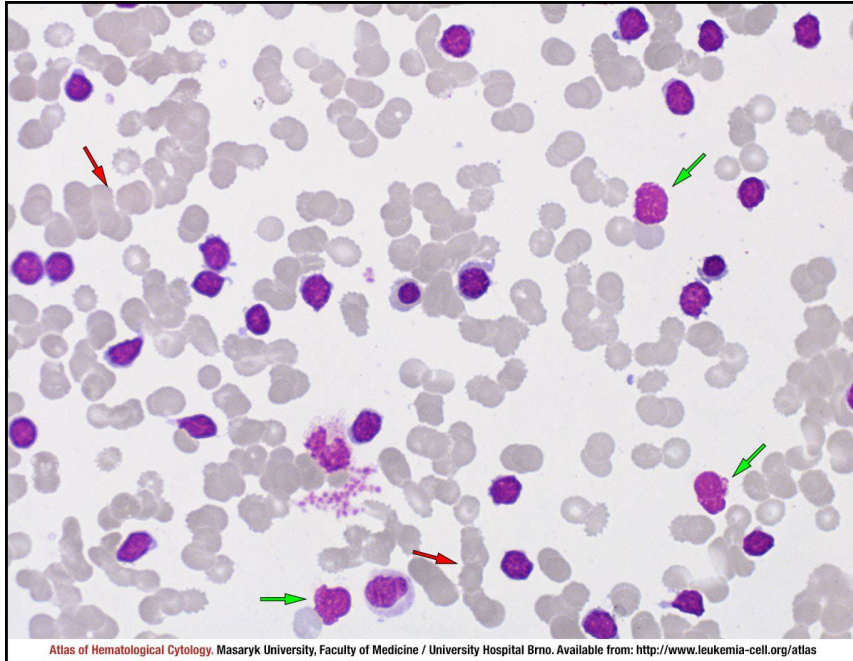
2cm below costal margin

What are the abnormalities?

- **Rising lymphocyte count for 6 months**
- **Smear cells**
- **Hb and plt normal**

Collect Date :	24/02/17	29/06/17	28/07/17	09/08/17	16/08/17		
Collect Time :	09:28	11:58	11:59	12:13	11:51		
Arrive Date :	24/02/17	29/06/17	28/07/17	09/08/17	16/08/17		
Arrive Time :	11:23	14:36	14:38	14:21	14:31		
Request No. :	H4040631	H4136705	H4160222	H4169678	H4175502	Reference	Units
Urgency :	--	--	--	--	--	Range	
Specimen type: EDTA blood							
COMPLETE BLOOD COUNT							
WBC	11.4 H	16.2 H	19.1 H	18.6 H	23.2 H	3.7 - 9.2	x10 ⁹ /L
RBC	4.59	4.30	4.29 L	4.28 L	4.38	4.30 - 5.90	x10 ¹² /L
HGB	14.8	14.0	13.9	14.0	13.9	13.4 - 17.1	g/dL
HCT	0.441	0.411	0.414	0.410	0.422	0.400-0.510	L/L
MCV	96.0	95.5	96.7	95.8	96.4	82.0 - 97.0	fL
MCH	32.3	32.5	32.3	32.6	31.8	27.0 - 33.0	pg
MCHC	33.6	34.0	33.5	34.1	33.0	32.0 - 35.0	g/dL
RDW	14.1 H	13.7	13.6	13.7	13.8	11.0 - 14.0	%
PLT	238	193	193	205	216	145 - 370	x10 ⁹ /L
MPV	8.0	8.2	8.6	8.7	8.6	7.0 - 10.0	fL
WBC DIFFERENTIAL							
Neutrophil	3.9	3.6	3.7	4.9	3.8	1.7 - 5.8	x10 ⁹ /L
NEU %	24.7 L	22.2 L	18.2 L	26.4 L	16.5 L	43.0 - 71.0	%
Lymphocyte	6.3 H	11.4 H	14.1 H	12.5 H	17.9 H	1.0 - 3.1	x10 ⁹ /L
LYM %	55.2 H	70.5 H	72.9 H	67.0 H	77.2 H	19.0 - 46.0	%
Monocyte	0.7	0.7	0.7	0.9 H	0.9 H	0.1 - 0.8	x10 ⁹ /L
MON %	5.8	4.5	3.8	4.7	3.9	3.0 - 12.0	%
Eosinophil	0.3	0.3	0.4	0.2	0.3	0.0 - 0.5	x10 ⁹ /L
EOS %	2.7	1.9	1.9	1.0	1.5	0 - 7.0	%
Basophil	0.1	0.1	0.2 H	0.2 H	0.2 H	0.0 - 0.1	x10 ⁹ /L
BAS %	1.0	0.8	1.2	0.9	0.9	0 - 1.4	%
RBC MORPHOLOGY							
RBC Morphology	normal						
ESR, automated	9				< 32		mm/hr
Remarks	Y	Y	Y	Y	Y		
Comment:							
17H416022	A few smear cells seen						
17H4169678	A few smear cells seen						
17H4175502	A few smear cells seen						

Chronic lymphocytic leukaemia



- Usually occurs in **older** patients
- Incidental finding of **lymphocytosis**
- **Constitutional symptoms:** fatigue, weight loss, lymphadenopathy, hepatosplenomegaly
- May **transform** to aggressive large cell lymphoma (Richter transformation)

Chronic lymphocytic leukaemia

- Treatment decision based on **clinical staging and symptoms**
- No treatment in early stages
- **Treatment**
 - **W&W**
 - **Young and fit:** chemotherapy based with fludarabine, cyclophosphamide, rituximab
 - **Old and frail:** chlorambucil
 - **New small molecules** targeting specific signal pathways:
 - Ibrutinib, Acalabrutinib (BTK inhibitor)
 - Idelalisib (PI3K inhibitor),
 - Venetoclax (BCL2 inhibitor)

Table 3. Rai classification and Binet staging systems for CLL

System	Clinical features	Median survival, y
Rai stage (simplified 3-stage)		
0 (low risk)	Lymphocytosis in blood and marrow only	> 10
I and II (intermediate risk)	Lymphadenopathy, splenomegaly +/- hepatomegaly	7
III and IV (high risk)	Anemia, thrombocytopenia	0.75-4
Binet group		
A	Fewer than 3 areas of lymphadenopathy; no anemia or thrombocytopenia	12
B	More than 3 involved node areas; no anemia or thrombocytopenia	7
C	Hemoglobin < 100 g/L platelets < 100 × 10 ⁹ g/L	2-4

Gribben. Blood. 2010; 115:187-197

© Thomas Mangelsen / Comedy Wildlife Photography Awards 2019



Bone Marrow Transplantation (BMT)

Haematopoietic Stem Cell
Transplantation (HSCT)

FAQ in HSCT

1. What are the indications for HSCT?
2. BMT vs HSCT?
3. How does HSCT cure leukaemia?
4. Who can be the donor for HSCT?

Indications for Allogeneic HSCT

- AML/ALL – high risk at first remission or at relapsed
- MDS – high risk
- CML (T315I, AP/BC)
- Transformed MPN
- Relapsed lymphoma
- Aplastic anemia
- Thalassemia major
- Sickle cell anemia
- Fanconi's anemia
- Blackfan-Diamond
- SCID
- Wiskott-Aldrich
- Inborn errors of metabolism

Allogeneic HSCT for leukaemia

- BMT = bone marrow transplantation
- PBSCT = peripheral blood stem cell transplantation

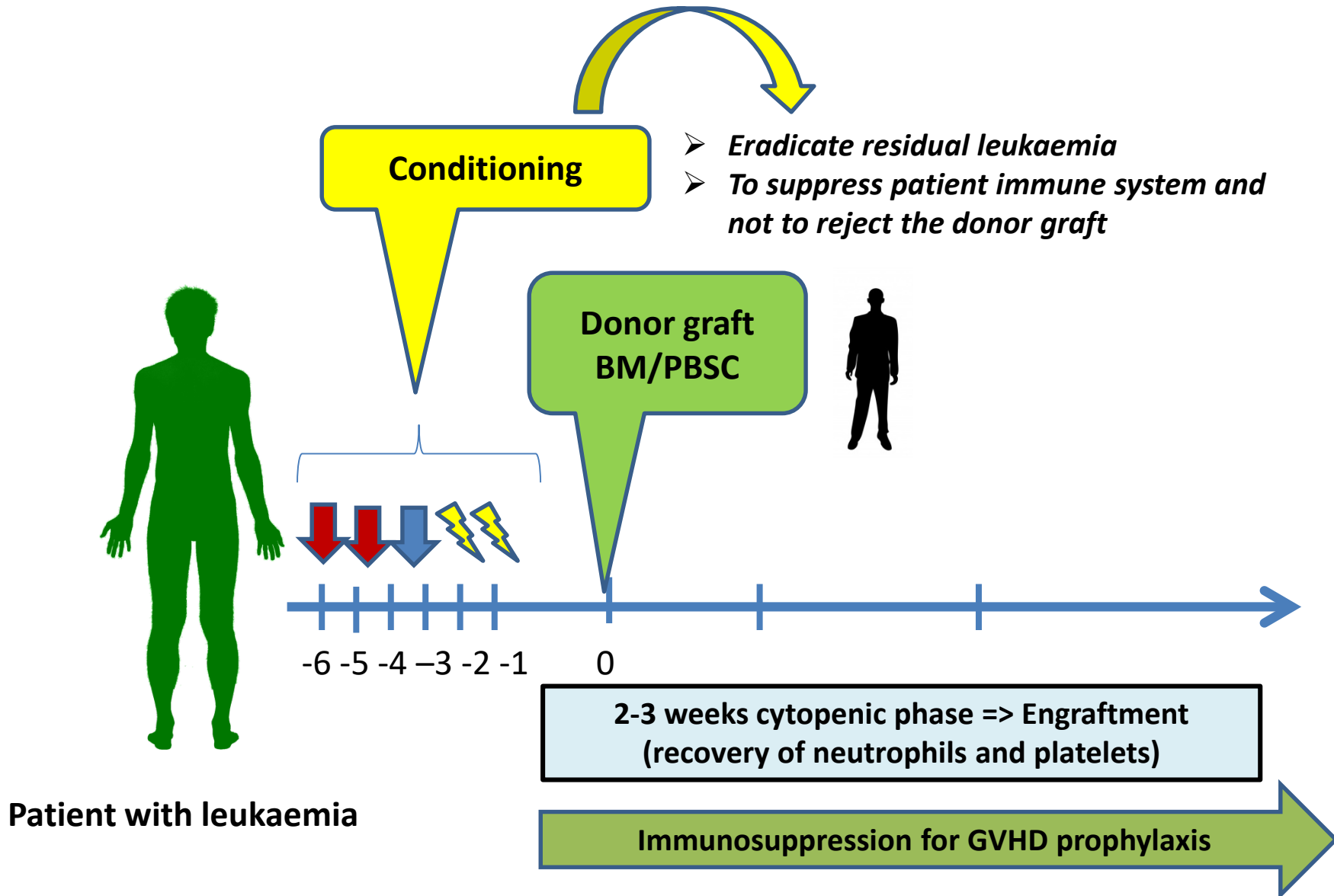


- HSCT = haematopoietic stem cell transplantation

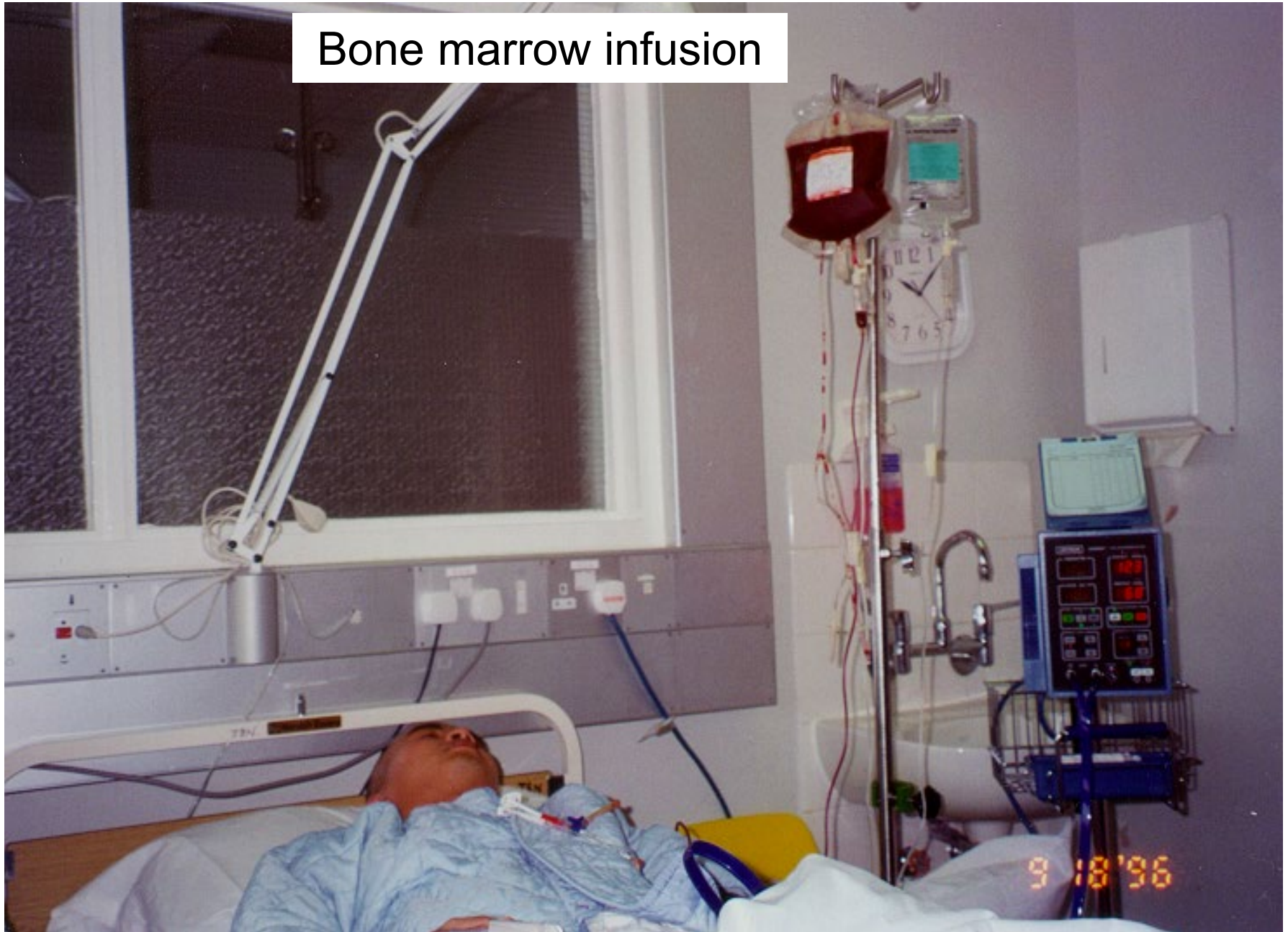
How does HSCT cure leukaemia?

- Use high dose **chemotherapy/radiotherapy** for leukaemia eradication
- **Replace** the diseased bone marrow with a healthy marrow from a normal donor
- **Graft versus Leukaemia (GVL) effect** -> lower relapse rate
- **HLA matched donor**
- HLA matched related siblings – first choice
- Other donors
- *ABO incompatibility is **not** an exclusion in HSCT*

How is Allo- HSCT performed ?



Bone marrow infusion

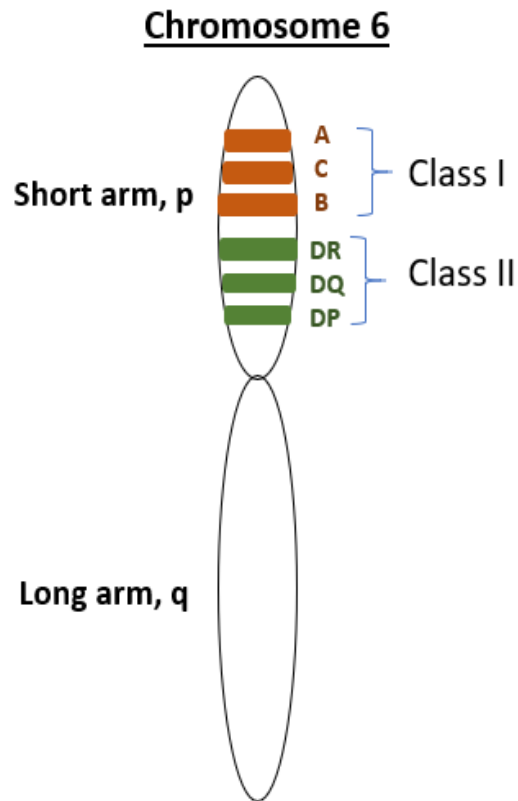


Who can be donor for HSCT?

Depending on donor type

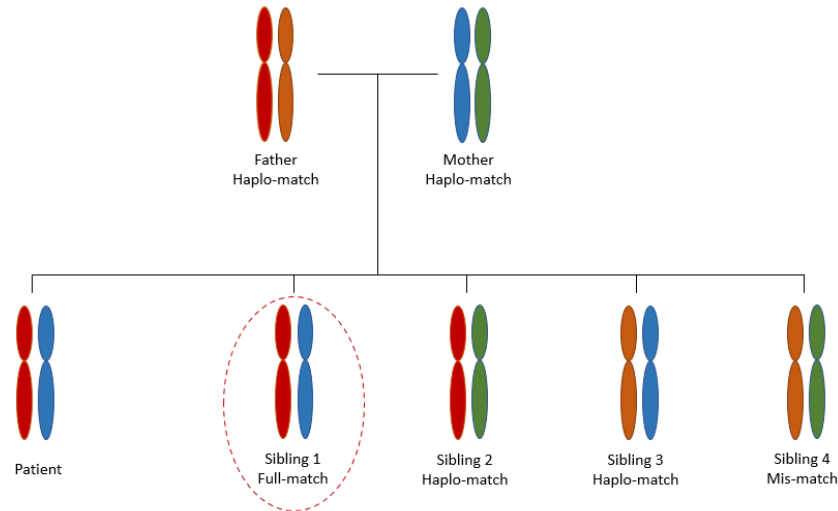
	Donor
Autologous	Self
Syngeneic	Monocytotic twins
Allogeneic	HLA matched sibling
	HLA matched unrelated donor
	HLA matched cord blood
	Haploidentical (mismatched family) donor

Human Leukocyte Antigen (HLA) System



- Encoded by the **Major Histocompatibility Complex (MHC)** gene on the short arm of chromosome 6
- The HLA antigens are expressed on cell surface and involved in recognition of “**self**” (**tolerate**) and “**non-self**” (**eliminate**).
 - **Class I** present on all nucleated cells
 - **Class II** present on immune cells
- HLA matching is crucial in HSCT to avoid graft rejection and graft versus host disease

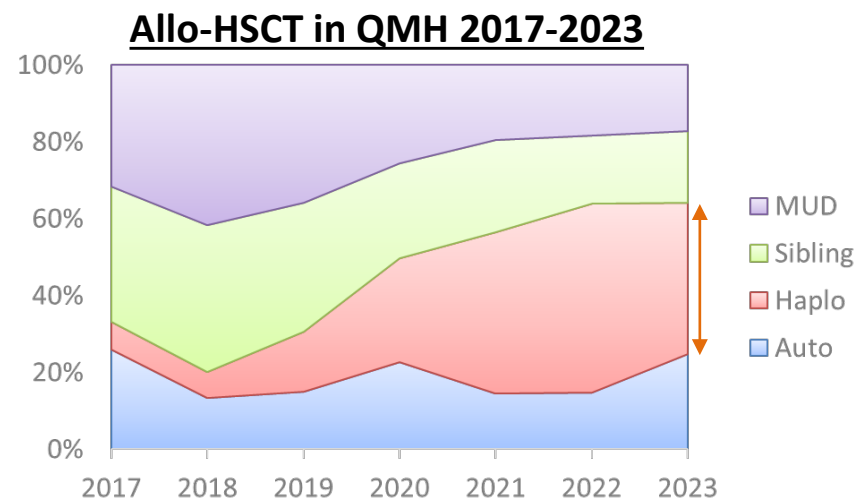
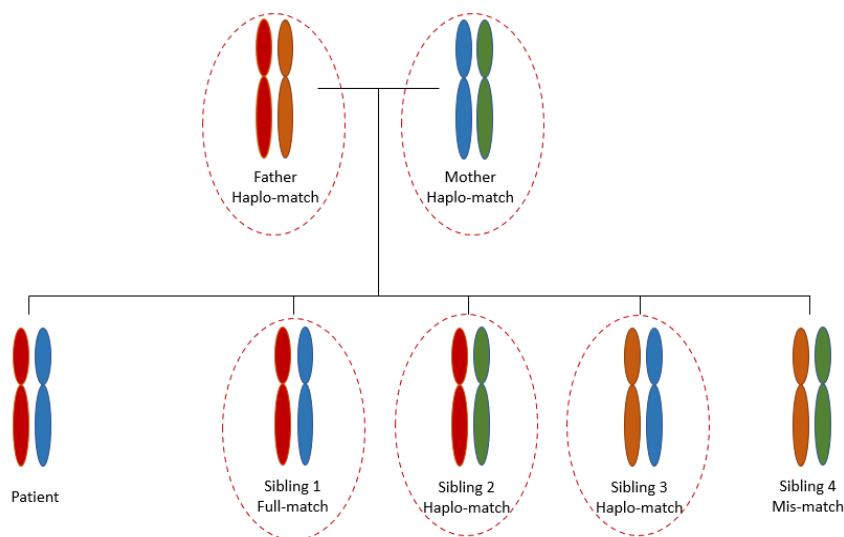
HLA compatibility within family



- The HLA genes are inherited from parents in a simple Mendelian co-dominant fashion
- Each sibling inherited one haplotype from each patient
- Each sibling has a 1 in 4 chance to be HLA identical to the patient
- HLA-identical by descent

If no HLA compatible siblings within family

1. **Matched unrelated donor** from local and overseas donor registries
2. **Cord blood** (uncommon in adults because of dose)
3. **Haplo-identical donor** (*mismatched family donor*)



Complications of HSCT

- Infection
 - Haemorrhage
 - Veno-occlusive disease of liver (VOD)
- } Related to high dose chemotherapy
- Graft-versus-host disease (GVHD)
 - Graft rejection (i.e. host-versus-graft)
- } In allogeneic HSCT
- Cataract (*mainly due to total body irradiation (TBI)*)
 - Immunodeficiency
 - Endocrine dysfunction & infertility
 - Secondary malignancy
- } Late effects
- Relapse of disease

Acute GVHD of Skin



Chronic GVHD





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Allogeneic Haematopoietic Stem Cell Transplantation

Dr Garret MK LEUNG

MBBS(HK), MRCP(UK), FHKAM(Medicine)
Associate Consultant

Dr Joycelyn PY SIM

MBBS(HK), FRCP(Edin), FHKAM(Medicine)
Consultant
Department of Medicine, Queen Mary Hospital



Dr Garret MK LEUNG

Dr Joycelyn PY SIM

INTRODUCTION

Allogeneic haematopoietic stem cell transplantation (allo-HSCT) is considered the only curative treatment of choice for many high-risk haematological malignancies. It can be seen as an effective form of immunotherapy applied systematically against leukaemia. In addition to the cytotoxic effect of the high-dose pre-transplantation conditioning chemotherapy, the donor-derived stem cells provide allo-immunity that enables a graft-versus-tumour (GVT) effect to eradicate residual disease and prevent relapse. Along with the introduction of the various novel agents, many would have expected allo-HSCT to become a "sunset industry". Instead, the annual number of allo-HSCT has continued to increase worldwide in the past decade.^{1,2} Here we shall review the development of allo-HSCT with emphasis on recent new advances and the current situation of allo-HSCT in Hong Kong.

TRENDS IN ALLO-HSCT IN THE PAST 50 YEARS

The first allo-HSCT was pioneered by the 1990 Nobel laureate, Dr E. Donnall Thomas, and reported in the New England Journal of Medicine on September 12, 1957. In the beginning, access to allo-HSCT was limited. This was due to the high transplant-related mortality (TRM) secondary to the high-dose conditioning chemotherapy, the risks of life-threatening infections and bleeding during the cytopenic phase, and the limited treatment strategies for severe graft-versus-host disease. Donor availability was another issue as only patients with a human leukocyte antigen (HLA)-matched sibling could undergo this treatment.

A comparison of transplantations performed over the last two decades would show that survival in allo-HSCT recipients have improved across all age spectra.³ There has been significant improvement in the field of transplantation over the last 50 years, attributable to the introduction of the less toxic reduced-intensity conditioning (RIC), the improved supportive care including more potent antimicrobial agents and better transfusion support, and the newly available FDA-approved drug treatments for both acute and chronic graft-versus-host disease (GVHD). The introduction of unrelated donor and cord blood transplantations, and the subsequent establishment of the international unrelated donor registries and cord blood banks have significantly increased donor availability and have allowed patients who do not have any HLA-matched sibling to benefit from this treatment. All these advancements have led to a dramatic increase in the number of allo-HSCT, especially among "silver hair" patients who were once considered

ineligible. According to the CIBMTR database, only 48 patients, representing 2% of all allo-HSCT performed in the year 2000, were aged > 65 years. In 2019, the number of allo-HSCT recipients aged > 65 years had grown to 1,888 patients, representing 26% of the total number of allo-HSCT performed in 2019. Given that the average age of diagnosis of acute myeloid leukaemia (AML), a blood malignancy which is the most common indication for allo-HSCT, is 68 years of age, this advancement has greatly improved the prognosis of the "silver hair" AML patients.⁴

WHAT IS NEW IN ALLO-HSCT OVER THE LAST DECADE?

Donor availability remains one of the major challenges to the success of allo-HSCT. Only about a quarter of the patients who need the transplant can find an HLA-identical sibling donor. Despite the expansion of the worldwide unrelated donor programme, the complicated search process for an unrelated donor necessitates 4 - 6 months of lag time from initiation of a search to the actual donation of stem cells. Unrelated cord blood as an alternative donor source offers the advantages of easy procurement and immediate availability; the low cell content poses engraftment problems for transplantation in adult patients.

WHY HAPLOIDENTICAL, THE EARLY FAILURES, THE BARRIERS, AND THE SOLUTIONS

In contrast, almost all patients who need an HSCT would have at least one identifiable haploidentical donor within his family, nuclear or extended. Biological children, parents, siblings, and frequently even more distant family members who share one haplotype are potentially qualified as donors (Fig. 1). In addition, these donors are often highly motivated and readily willing to adjust their own life plans in order to accommodate to patients' transplant schedule and changes in clinical conditions.

Yet, because of the increased T-cell mediated allo-reactivity, the early development of haploidentical HSCT (haplo-HSCT) was hindered by the high rates of GVHD and graft failure, resulting in ~10% long-term survival.⁵

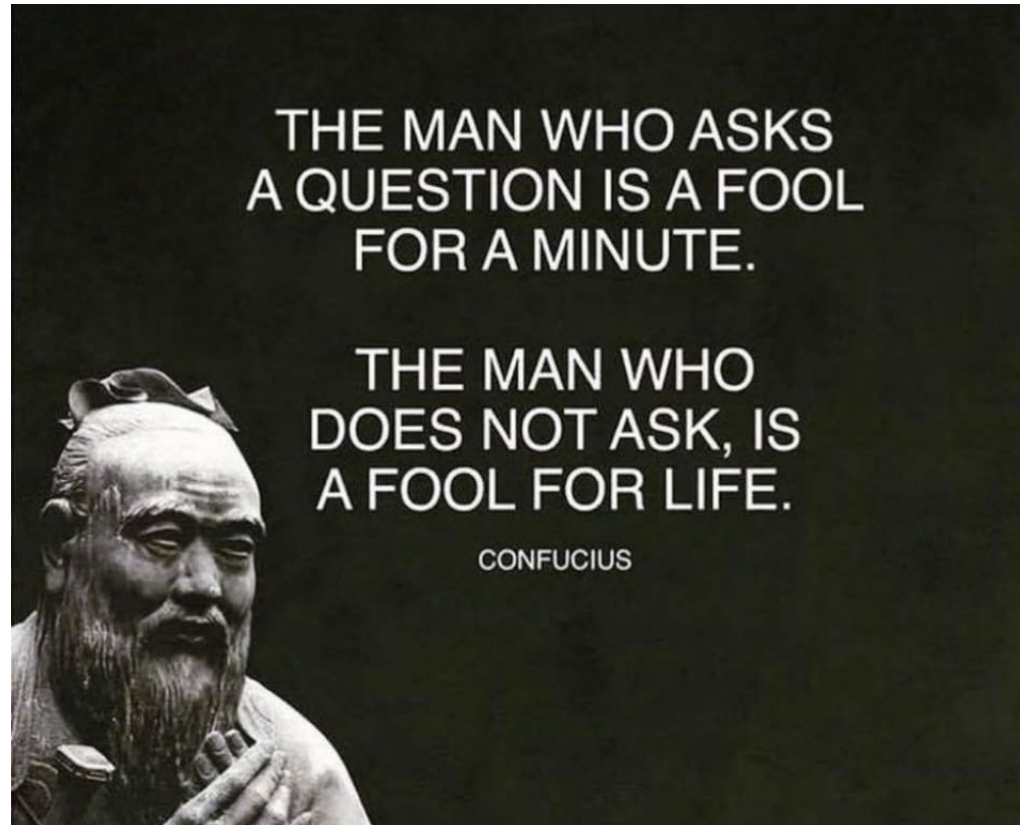
By removing the T-cells from the graft, Reisner and colleagues performed the first successful haplo-HSCT in children with severe combined immunodeficiency (SCID) using T-cell depleted (TCD) haploidentical

Take home message

High white cell count

- **Reactive** causes are common in clinical practice
 - Ascertainment of the abnormal **cell type**
 - Leukaemia treatment depends on specific **diagnosis, disease risk and patient comorbidities**
 - Look out for **haematological emergencies**
-
- **Allogeneic HSCT** for high risk cases and at relapse
 - **HLA-matching** is essential in allogeneic HSCT

THE END
QUESTIONS ?



THE MAN WHO ASKS
A QUESTION IS A FOOL
FOR A MINUTE.

THE MAN WHO
DOES NOT ASK, IS
A FOOL FOR LIFE.

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